

Rex Chen

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Education

CARNEGIE MELLON UNIVERSITY

School of Computer Science

PhD in Societal Computing (GPA: 4.25/4.33)

Thesis: [*Rethinking the Design of Human-Centric AI Systems for Deployment in Transportation*](#)

Aug 2020 – Aug 2025

Advisors: Fei Fang,

Norman Sadeh

UNIVERSITY OF BRITISH COLUMBIA

Department of Computer Science

Honours BSc in Computer Science (Average: 92.9%)

Sep 2015 – May 2020

Advisor: Kevin Leyton-Brown

Selected Conference Papers

* Equal contribution

- [1] **Rex Chen**, Stephanie Milani, Zhicheng Zhang, Fei Fang, & Norman Sadeh (2025) “[Making Teams and Influencing Agents: Efficiently Coordinating Decision Trees for Interpretable Multi-Agent Reinforcement Learning](#)”. *Proceedings of the 2025 AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES '25)*, pp. 1–10.
- [2] **Rex Chen**, Karen Wu, John McCartney, Fei Fang, & Norman Sadeh (2025) “[Out of the Past: An AI-Enabled Pipeline for Traffic Simulation from Noisy, Multimodal Detector Data and Stakeholder Feedback](#)”. *Proceedings of the 2025 Winter Simulation Conference (WSC '25)*, pp. 1–12.
- [3] **Rex Chen**, Ruiyi Wang, Fei Fang, & Norman Sadeh (2025) “[Missing Pieces: How Do Designs that Expose Uncertainty Longitudinally Impact Trust in AI Decision Aids? An In Situ Study of Gig Drivers](#)”. *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)*, pp. 1–27.
- [4] **Rex Chen**, Kathleen M. Carley, Fei Fang, & Norman Sadeh (2023) “[Purpose in the Machine: Do Traffic Simulators Produce Distributionally Equivalent Outcomes for Reinforcement Learning Applications?](#)”. *Proceedings of the 2023 Winter Simulation Conference (WSC '23)*, pp. 1–12.
- [5] **Rex Chen**, Fei Fang, & Norman Sadeh (2022) “[The Real Deal: A Review of Challenges and Opportunities in Moving Reinforcement Learning-Based Traffic Signal Control Systems Towards Reality](#)”. *12th International Workshop on Agents in Traffic and Transportation (ATT '22 @ IJCAI '22)*, *CEUR Workshop Proceedings* **3173**: 14–31.
- [6] **Rex Chen**, Fei Fang, Aleecia M. McDonald, Thomas Norton, & Norman Sadeh (2021) “[Fighting the Fog: Evaluating the Clarity of Privacy Disclosures in the Age of CCPA](#)”. *Proceedings of the 20th Workshop on Privacy in the Electronic Society (WPES '21)*, pp. 73–102.
- [7] Chris Cameron*, **Rex Chen***, Jason Hartford*, & Kevin Leyton-Brown (2020) “[Predicting Propositional Satisfiability via End-to-End Learning](#)”. *Proceedings of the 2020 AAAI Conference on Artificial Intelligence (AAAI '20)*, **34**(04): 3324–3331.

Skills

- **Languages/Frameworks:** Python (PyTorch, TensorFlow, Django, Flask), R, SQL (Hive, Spark, PostgreSQL, MySQL), C#, Docker, HTML, CSS, JavaScript, Bash, PowerShell, C++, Java, MATLAB
- **Knowledge Areas:** Reinforcement learning, transportation research, interpretable machine learning, human-computer interaction, statistical modelling, causal inference, computational game theory

Industrial Work Experience

DATA SCIENTIST INTERN

Lyft

May 2024 – Aug 2024

- Designed, trained, and productionised ensemble machine learning models to predict the responses of gig drivers to real-time incentive mechanisms across multiple markets.
- Applied causal inference methods to derive actionable insights for setting incentive policies, including a proposed feature envisioned in collaboration with multiple cross-functional teams.

SOFTWARE DEVELOPER CO-OP

Change Healthcare

Sep 2017 – Apr 2018

- Worked with senior software developers to code, test, and deploy bug fixes and upgrades for two leading healthcare workflow products.
- Took on a primary role in researching, developing, and integrating an authentication service for inter-service communications using an open-source library in the .NET Core framework.

Academic Work Experience

PHD STUDENT

Software & Societal Systems Department, Carnegie Mellon University Aug 2020 – Aug 2025

- Researching applications of multi-agent reinforcement learning, human-computer interaction, and computational game theory to problems in transportation, including traffic signal control and ridesharing.
- Led collaborative research projects involving three undergraduate and five master's students, entailing various research methods such as machine learning experiments, user studies, and literature reviews.

Awards & Honours

- 2023: **Presidential Graduate Fellowship**, awarded by the CMU School of Computer Science
- 2023: **NSERC Postgraduate Scholarship – Doctoral** for proposal “Large Scale Learning for Multi-Agent Communication & Coordination in Transportation”
- 2022: **NSF Graduate Research Fellowship – Honourable Mention** for proposal “Large Scale Learning for Multi-Agent Communication & Coordination in Transportation”
- 2022: **Mobility21** (USDOT/CMU National University Transportation Centre) funding for proposal “Alleviating Traffic Congestion: Developing and Evaluating Novel Multi-Agent Reinforcement Learning Traffic Light Coordination Techniques” (PI: Fei Fang; Co-PI: Norman Sadeh)
- 2021: **Tang Family Endowed Innovation Fund** for proposal “Large Scale Learning for Multi-Agent Communication & Coordination in Transportation” (PI: Fei Fang)
- 2018, 2019: **NSERC Undergraduate Student Research Awards**